

Status of Petrale sole stock off the U.S. West Coast in 2023



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1 Data

1.1 Life History

1.2 Catch

1.3 Indices and Standardization

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1.5 Absolute Abundance

1.6 Environmental/Ecosystem Indicator Data

2 Assessment

2.1 Current Modeling Approach

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2.3 Bridging

2.4 Modeling Results

2.4.1 Parameter Estimates

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2.4.4 Model Diagnostics

2.5 Sensitivity Analyses

2.6 Management Benchmarks

2.7 Projections

2.8 Harvest Control Rules

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3 Discussion

4 Acknowledgements

This document was produced using the R package asar (Schiano et al. 2026), which is free to use and publicly available on [GitHub](#).

5 References

Schiano, S., Breitbart, S., and Saul, S. 2026. Asar: Build NOAA stock assessment report. Available from <https://github.com/nmfs-ost/asar>.

6 Tables

Table 1: Calculated index of abundance and corresponding CVs for the fleets and surveys identified in the column headers.

Year	Index Observed Triennial (se)	Index Predicted Triennial (se)	Index Observed Wcgbts (se)	Index Predicted Wcgbts (se)
1980	1,835.00 (0.38)	3,322.83 (0.38)	-	-
1983	1,968.45 (0.34)	2,438.46 (0.34)	-	-
1986	1,819.10 (0.34)	2,211.88 (0.34)	-	-
1989	2,590.42 (0.34)	2,177.91 (0.34)	-	-
1992	1,860.61 (0.34)	1,933.47 (0.34)	-	-
1995	2,071.20 (0.35)	2,516.70 (0.35)	-	-
1998	3,071.65 (0.33)	2,784.97 (0.33)	-	-
2001	3,411.62 (0.34)	3,298.33 (0.34)	-	-
2004	8,975.17 (0.34)	4,051.63 (0.34)	22,772.20 (0.09)	20,790.60 (0.09)
2003	-	-	18,526.50 (0.09)	19,373.70 (0.09)
2005	-	-	23,426.80 (0.08)	20,698.20 (0.08)
2006	-	-	21,032.50 (0.08)	19,167.60 (0.08)
2007	-	-	18,081.30 (0.08)	17,844.10 (0.08)
2008	-	-	16,272.40 (0.08)	17,074.90 (0.08)
2009	-	-	16,343.10 (0.08)	17,896.00 (0.08)
2010	-	-	23,898.40 (0.07)	22,500.50 (0.07)
2011	-	-	31,344.10 (0.07)	30,760.20 (0.07)
2012	-	-	36,896.00 (0.07)	40,647.60 (0.07)
2013	-	-	55,341.80 (0.09)	49,036.70 (0.09)
2014	-	-	57,060.90 (0.07)	54,277.90 (0.07)
2015	-	-	51,353.50 (0.07)	56,914.90 (0.07)
2016	-	-	59,009.80 (0.07)	57,499.30 (0.07)
2017	-	-	60,348.00 (0.07)	56,575.70 (0.07)
2018	-	-	42,010.00 (0.08)	54,735.80 (0.08)
2019	-	-	43,739.20 (0.1)	52,972.80 (0.1)
2021	-	-	50,087.90 (0.07)	49,724.90 (0.07)
2022	-	-	51,429.80 (0.08)	45,694.10 (0.08)

Table 2: Landed catch by fleet and year in mt. (1 of 6)

Year	Landings Observed Weight North (mt) (se)	Landings Predicted Weight North (mt) (se)	Landings Observed Weight South (mt) (se)	Landings Predicted Weight South (mt) (se)
1876	0.00 (0.01)	0.00 (0.01)	1.00 (0.01)	1.00 (0.01)
1877	0.00 (0.01)	0.00 (0.01)	1.00 (0.01)	1.00 (0.01)
1878	0.00 (0.01)	0.00 (0.01)	1.00 (0.01)	1.00 (0.01)
1879	0.00 (0.01)	0.00 (0.01)	1.00 (0.01)	1.00 (0.01)
1880	0.00 (0.01)	0.00 (0.01)	11.55 (0.01)	11.55 (0.01)
1881	0.00 (0.01)	0.00 (0.01)	22.10 (0.01)	22.10 (0.01)
1882	0.00 (0.01)	0.00 (0.01)	32.65 (0.01)	32.65 (0.01)
1883	0.00 (0.01)	0.00 (0.01)	43.20 (0.01)	43.20 (0.01)
1884	0.00 (0.01)	0.00 (0.01)	53.75 (0.01)	53.75 (0.01)
1885	0.00 (0.01)	0.00 (0.01)	64.30 (0.01)	64.30 (0.01)
1886	0.00 (0.01)	0.00 (0.01)	74.85 (0.01)	74.85 (0.01)
1887	0.00 (0.01)	0.00 (0.01)	85.40 (0.01)	85.40 (0.01)
1888	0.00 (0.01)	0.00 (0.01)	95.95 (0.01)	95.95 (0.01)
1889	0.00 (0.01)	0.00 (0.01)	106.50 (0.01)	106.50 (0.01)
1890	0.00 (0.01)	0.00 (0.01)	117.05 (0.01)	117.05 (0.01)
1891	0.00 (0.01)	0.00 (0.01)	127.60 (0.01)	127.60 (0.01)
1892	0.00 (0.01)	0.00 (0.01)	138.15 (0.01)	138.15 (0.01)
1893	0.00 (0.01)	0.00 (0.01)	148.71 (0.01)	148.71 (0.01)
1894	0.00 (0.01)	0.00 (0.01)	159.26 (0.01)	159.26 (0.01)
1895	0.00 (0.01)	0.00 (0.01)	169.81 (0.01)	169.81 (0.01)
1896	0.24 (0.01)	0.24 (0.01)	180.36 (0.01)	180.36 (0.01)
1897	0.20 (0.01)	0.20 (0.01)	190.91 (0.01)	190.91 (0.01)
1898	0.15 (0.01)	0.15 (0.01)	201.46 (0.01)	201.46 (0.01)
1899	0.15 (0.01)	0.15 (0.01)	212.01 (0.01)	212.01 (0.01)
1900	0.15 (0.01)	0.15 (0.01)	222.56 (0.01)	222.56 (0.01)
1901	0.14 (0.01)	0.14 (0.01)	233.11 (0.01)	233.11 (0.01)
1902	0.14 (0.01)	0.14 (0.01)	243.66 (0.01)	243.66 (0.01)
1903	0.13 (0.01)	0.13 (0.01)	254.21 (0.01)	254.21 (0.01)

Table 3: Landed catch by fleet and year in mt. (2 of 6)

Year	Landings Observed Weight North (mt) (se)	Landings Predicted Weight North (mt) (se)	Landings Observed Weight South (mt) (se)	Landings Predicted Weight South (mt) (se)
1904	0.13 (0.01)	0.13 (0.01)	264.76 (0.01)	264.76 (0.01)
1905	0.13 (0.01)	0.13 (0.01)	275.31 (0.01)	275.31 (0.01)
1906	0.12 (0.01)	0.12 (0.01)	285.86 (0.01)	285.86 (0.01)
1907	0.12 (0.01)	0.12 (0.01)	296.41 (0.01)	296.41 (0.01)
1908	0.11 (0.01)	0.11 (0.01)	306.96 (0.01)	306.96 (0.01)
1909	0.11 (0.01)	0.11 (0.01)	317.51 (0.01)	317.51 (0.01)
1910	0.10 (0.01)	0.10 (0.01)	328.06 (0.01)	328.06 (0.01)
1911	0.10 (0.01)	0.10 (0.01)	338.61 (0.01)	338.61 (0.01)
1912	0.10 (0.01)	0.10 (0.01)	349.16 (0.01)	349.16 (0.01)
1913	0.09 (0.01)	0.09 (0.01)	359.71 (0.01)	359.71 (0.01)
1914	0.09 (0.01)	0.09 (0.01)	370.26 (0.01)	370.26 (0.01)
1915	0.08 (0.01)	0.08 (0.01)	380.81 (0.01)	380.81 (0.01)
1916	0.08 (0.01)	0.08 (0.01)	386.42 (0.01)	386.42 (0.01)
1917	0.08 (0.01)	0.08 (0.01)	526.41 (0.01)	526.41 (0.01)
1918	0.07 (0.01)	0.07 (0.01)	423.85 (0.01)	423.85 (0.01)
1919	0.07 (0.01)	0.07 (0.01)	333.44 (0.01)	333.44 (0.01)
1920	0.06 (0.01)	0.06 (0.01)	230.49 (0.01)	230.49 (0.01)
1921	0.06 (0.01)	0.06 (0.01)	293.76 (0.01)	293.76 (0.01)
1922	0.05 (0.01)	0.05 (0.01)	424.78 (0.01)	424.78 (0.01)
1923	0.05 (0.01)	0.05 (0.01)	427.36 (0.01)	427.36 (0.01)
1924	0.05 (0.01)	0.05 (0.01)	532.86 (0.01)	532.86 (0.01)
1925	0.04 (0.01)	0.04 (0.01)	528.47 (0.01)	528.47 (0.01)
1926	0.04 (0.01)	0.04 (0.01)	521.67 (0.01)	521.67 (0.01)
1927	0.04 (0.01)	0.04 (0.01)	632.04 (0.01)	632.04 (0.01)
1928	0.00 (0.01)	0.00 (0.01)	620.09 (0.01)	620.09 (0.01)
1929	1.54 (0.01)	1.54 (0.01)	706.04 (0.01)	706.04 (0.01)
1930	1.22 (0.01)	1.22 (0.01)	658.83 (0.01)	658.83 (0.01)
1931	17.17 (0.01)	17.17 (0.01)	623.93 (0.01)	623.93 (0.01)

Table 4: Landed catch by fleet and year in mt. (3 of 6)

Year	Landings Observed Weight North (mt) (se)	Landings Predicted Weight North (mt) (se)	Landings Observed Weight South (mt) (se)	Landings Predicted Weight South (mt) (se)
1932	41.73 (0.01)	41.73 (0.01)	556.66 (0.01)	556.66 (0.01)
1933	54.42 (0.01)	54.42 (0.01)	424.32 (0.01)	424.32 (0.01)
1934	68.97 (0.01)	68.97 (0.01)	1,110.98 (0.01)	1,110.98 (0.01)
1935	86.80 (0.01)	86.80 (0.01)	895.44 (0.01)	895.44 (0.01)
1936	115.33 (0.01)	115.33 (0.01)	510.98 (0.01)	510.98 (0.01)
1937	198.06 (0.01)	198.06 (0.01)	817.70 (0.01)	817.70 (0.01)
1938	134.64 (0.01)	134.64 (0.01)	989.10 (0.01)	989.10 (0.01)
1939	374.27 (0.01)	374.27 (0.01)	1,160.51 (0.01)	1,160.51 (0.01)
1940	602.24 (0.01)	602.24 (0.01)	714.63 (0.01)	714.63 (0.01)
1941	771.51 (0.01)	771.51 (0.01)	405.25 (0.01)	405.25 (0.01)
1942	1,928.65 (0.01)	1,928.65 (0.01)	277.41 (0.01)	277.41 (0.01)
1943	1,959.02 (0.01)	1,959.02 (0.01)	416.86 (0.01)	416.86 (0.01)
1944	1,111.90 (0.01)	1,111.90 (0.01)	509.50 (0.01)	509.50 (0.01)
1945	973.80 (0.01)	973.80 (0.01)	534.46 (0.01)	534.46 (0.01)
1946	1,596.12 (0.01)	1,596.12 (0.01)	1,209.41 (0.01)	1,209.41 (0.01)
1947	885.76 (0.01)	885.76 (0.01)	1,336.82 (0.01)	1,336.82 (0.01)
1948	1,971.81 (0.01)	1,971.81 (0.01)	2,308.64 (0.01)	2,308.64 (0.01)
1949	822.94 (0.01)	822.94 (0.01)	2,246.26 (0.01)	2,246.26 (0.01)
1950	1,480.79 (0.01)	1,480.79 (0.01)	1,980.46 (0.01)	1,980.46 (0.01)
1951	1,084.95 (0.01)	1,084.95 (0.01)	1,238.49 (0.01)	1,238.49 (0.01)
1952	864.21 (0.01)	864.21 (0.01)	1,312.51 (0.01)	1,312.51 (0.01)
1953	414.09 (0.01)	414.09 (0.01)	1,514.08 (0.01)	1,514.08 (0.01)
1954	634.02 (0.01)	634.02 (0.01)	1,891.97 (0.01)	1,891.97 (0.01)
1955	652.60 (0.01)	652.60 (0.01)	1,641.72 (0.01)	1,641.72 (0.01)
1956	571.36 (0.01)	571.36 (0.01)	1,277.89 (0.01)	1,277.89 (0.01)
1957	1,051.75 (0.01)	1,051.75 (0.01)	1,587.99 (0.01)	1,587.99 (0.01)
1958	890.87 (0.01)	890.87 (0.01)	1,427.80 (0.01)	1,427.80 (0.01)
1959	686.92 (0.01)	686.92 (0.01)	1,194.06 (0.01)	1,194.06 (0.01)

Table 5: Landed catch by fleet and year in mt. (4 of 6)

Year	Landings Observed Weight North (mt) (se)	Landings Predicted Weight North (mt) (se)	Landings Observed Weight South (mt) (se)	Landings Predicted Weight South (mt) (se)
1960	1,034.52 (0.01)	1,034.52 (0.01)	1,122.94 (0.01)	1,122.94 (0.01)
1961	1,010.41 (0.01)	1,010.41 (0.01)	1,538.01 (0.01)	1,538.01 (0.01)
1962	1,311.39 (0.01)	1,311.39 (0.01)	1,379.45 (0.01)	1,379.45 (0.01)
1963	1,205.57 (0.01)	1,205.57 (0.01)	1,504.99 (0.01)	1,504.99 (0.01)
1964	927.02 (0.01)	927.02 (0.01)	1,223.64 (0.01)	1,223.64 (0.01)
1965	901.65 (0.01)	901.65 (0.01)	1,207.58 (0.01)	1,207.58 (0.01)
1966	961.76 (0.01)	961.76 (0.01)	1,327.75 (0.01)	1,327.75 (0.01)
1967	925.63 (0.01)	925.63 (0.01)	1,255.79 (0.01)	1,255.79 (0.01)
1968	817.32 (0.01)	817.32 (0.01)	1,303.81 (0.01)	1,303.81 (0.01)
1969	948.66 (0.01)	948.66 (0.01)	1,300.34 (0.01)	1,300.34 (0.01)
1970	1,164.62 (0.01)	1,164.62 (0.01)	1,549.34 (0.01)	1,549.34 (0.01)
1971	1,314.18 (0.01)	1,314.18 (0.01)	1,573.50 (0.01)	1,573.50 (0.01)
1972	1,331.65 (0.01)	1,331.65 (0.01)	1,621.70 (0.01)	1,621.70 (0.01)
1973	1,645.58 (0.01)	1,645.58 (0.01)	1,304.98 (0.01)	1,304.98 (0.01)
1974	2,068.70 (0.01)	2,068.70 (0.01)	1,556.13 (0.01)	1,556.13 (0.01)
1975	2,006.61 (0.01)	2,006.61 (0.01)	1,483.24 (0.01)	1,483.24 (0.01)
1976	1,413.47 (0.01)	1,413.47 (0.01)	1,350.59 (0.01)	1,350.59 (0.01)
1977	1,269.72 (0.01)	1,269.72 (0.01)	998.22 (0.01)	998.22 (0.01)
1978	1,864.01 (0.01)	1,864.01 (0.01)	1,194.78 (0.01)	1,194.78 (0.01)
1979	1,666.00 (0.01)	1,666.00 (0.01)	1,388.97 (0.01)	1,388.97 (0.01)
1980	1,401.60 (0.01)	1,401.60 (0.01)	1,066.00 (0.01)	1,066.00 (0.01)
1981	1,236.04 (0.01)	1,236.05 (0.01)	805.15 (0.01)	805.15 (0.01)
1982	1,838.53 (0.01)	1,838.53 (0.01)	791.80 (0.01)	791.80 (0.01)
1983	1,630.29 (0.01)	1,630.30 (0.01)	583.90 (0.01)	583.90 (0.01)
1984	1,148.63 (0.01)	1,148.63 (0.01)	590.54 (0.01)	590.54 (0.01)
1985	982.89 (0.01)	982.89 (0.01)	856.56 (0.01)	856.56 (0.01)
1986	1,024.00 (0.01)	1,024.00 (0.01)	726.43 (0.01)	726.43 (0.01)
1987	1,381.18 (0.01)	1,381.20 (0.01)	823.66 (0.01)	823.67 (0.01)

Table 6: Landed catch by fleet and year in mt. (5 of 6)

Year	Landings Observed Weight North (mt) (se)	Landings Predicted Weight North (mt) (se)	Landings Observed Weight South (mt) (se)	Landings Predicted Weight South (mt) (se)
1988	1,354.06 (0.01)	1,354.07 (0.01)	795.12 (0.01)	795.12 (0.01)
1989	1,312.07 (0.01)	1,312.08 (0.01)	840.63 (0.01)	840.63 (0.01)
1990	1,086.22 (0.01)	1,086.22 (0.01)	678.42 (0.01)	678.42 (0.01)
1991	1,192.57 (0.01)	1,192.58 (0.01)	734.80 (0.01)	734.80 (0.01)
1992	1,021.46 (0.01)	1,021.46 (0.01)	532.04 (0.01)	532.04 (0.01)
1993	1,039.69 (0.01)	1,039.69 (0.01)	463.52 (0.01)	463.52 (0.01)
1994	825.58 (0.01)	825.58 (0.01)	549.56 (0.01)	549.56 (0.01)
1995	1,066.33 (0.01)	1,066.33 (0.01)	592.72 (0.01)	592.72 (0.01)
1996	1,010.33 (0.01)	1,010.33 (0.01)	818.27 (0.01)	818.27 (0.01)
1997	1,113.93 (0.01)	1,113.93 (0.01)	833.79 (0.01)	833.79 (0.01)
1998	989.95 (0.01)	989.95 (0.01)	472.82 (0.01)	472.82 (0.01)
1999	931.30 (0.01)	931.30 (0.01)	565.81 (0.01)	565.81 (0.01)
2000	1,253.23 (0.01)	1,253.23 (0.01)	640.04 (0.01)	640.04 (0.01)
2001	1,270.22 (0.01)	1,270.22 (0.01)	575.00 (0.01)	575.00 (0.01)
2002	1,317.36 (0.01)	1,317.36 (0.01)	479.73 (0.01)	479.73 (0.01)
2003	1,661.17 (0.01)	1,661.17 (0.01)	408.41 (0.01)	408.41 (0.01)
2004	1,471.05 (0.01)	1,471.05 (0.01)	492.92 (0.01)	492.92 (0.01)
2005	1,970.53 (0.01)	1,970.53 (0.01)	763.83 (0.01)	763.83 (0.01)
2006	1,858.41 (0.01)	1,858.41 (0.01)	751.79 (0.01)	751.79 (0.01)
2007	1,333.54 (0.01)	1,333.54 (0.01)	919.30 (0.01)	919.30 (0.01)
2008	1,294.91 (0.01)	1,294.91 (0.01)	924.80 (0.01)	924.80 (0.01)
2009	1,236.74 (0.01)	1,236.74 (0.01)	530.70 (0.01)	530.70 (0.01)
2010	589.72 (0.01)	589.72 (0.01)	213.38 (0.01)	213.38 (0.01)
2011	757.01 (0.01)	757.01 (0.01)	177.50 (0.01)	177.50 (0.01)
2012	896.27 (0.01)	896.27 (0.01)	221.45 (0.01)	221.45 (0.01)
2013	1,776.22 (0.01)	1,776.22 (0.01)	477.14 (0.01)	477.14 (0.01)
2014	1,783.41 (0.01)	1,783.41 (0.01)	625.33 (0.01)	625.33 (0.01)
2015	2,085.62 (0.01)	2,085.62 (0.01)	579.55 (0.01)	579.55 (0.01)

Table 7: Landed catch by fleet and year in mt. (6 of 6)

Year	Landings Observed Weight North (mt) (se)	Landings Predicted Weight North (mt) (se)	Landings Observed Weight South (mt) (se)	Landings Predicted Weight South (mt) (se)
2016	2,254.21 (0.01)	2,254.21 (0.01)	473.42 (0.01)	473.42 (0.01)
2017	2,313.91 (0.01)	2,313.91 (0.01)	616.71 (0.01)	616.71 (0.01)
2018	2,284.80 (0.01)	2,284.80 (0.01)	609.64 (0.01)	609.64 (0.01)
2019	2,079.95 (0.01)	2,079.95 (0.01)	536.96 (0.01)	536.96 (0.01)
2020	1,548.72 (0.01)	1,548.72 (0.01)	543.41 (0.01)	543.41 (0.01)
2021	2,103.03 (0.01)	2,103.03 (0.01)	776.08 (0.01)	776.08 (0.01)
2022	2,093.58 (0.01)	2,093.58 (0.01)	966.36 (0.01)	966.36 (0.01)

7 Figures

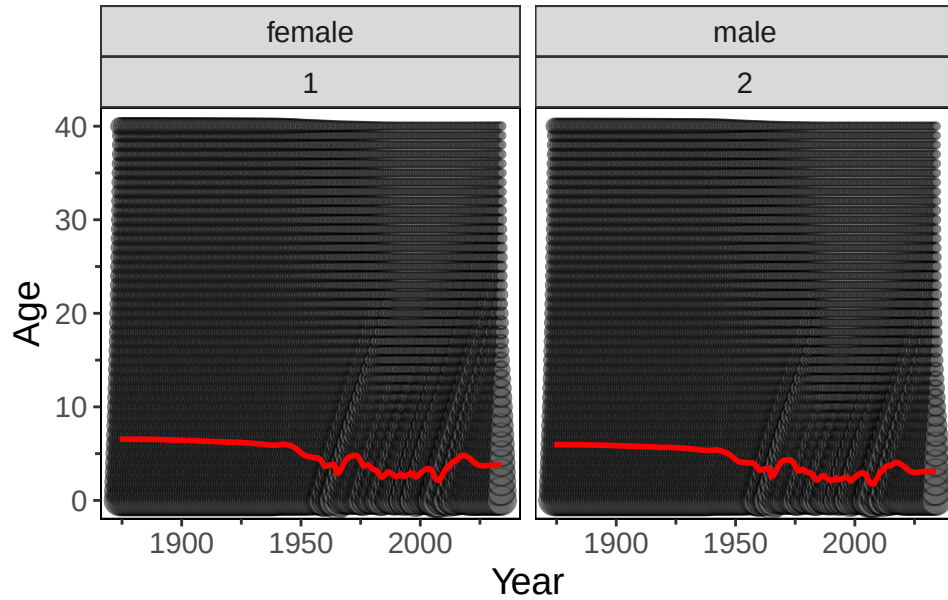


Figure 1: Model estimate of population numbers at age over time. The relative size of each bubble for a given year and age indicates the relative abundance in that category compared with others.

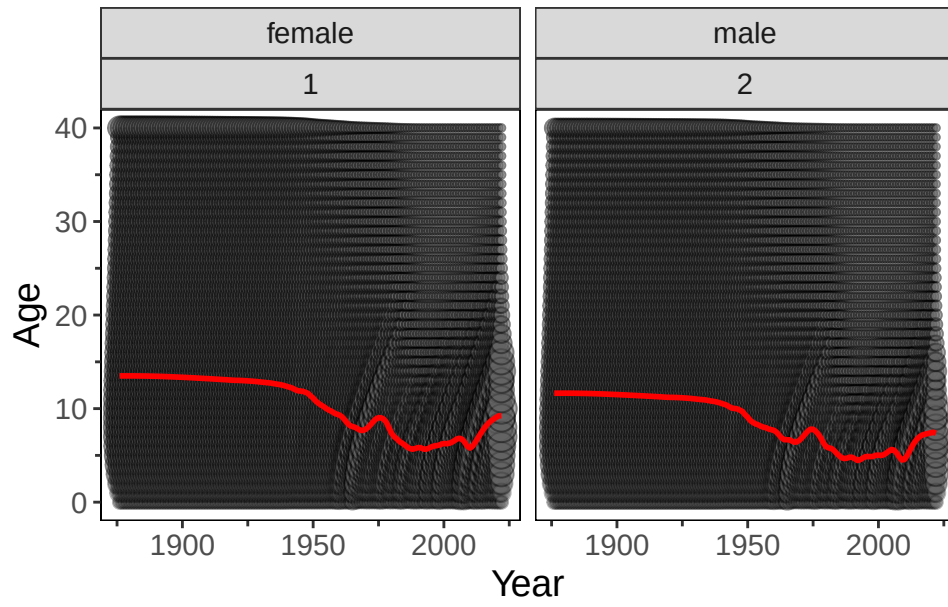


Figure 2: Estimated population numbers at age and population biomass at age over time. The relative size of each bubble for a given year and age indicates the relative abundance or biomass in that category compared with others.

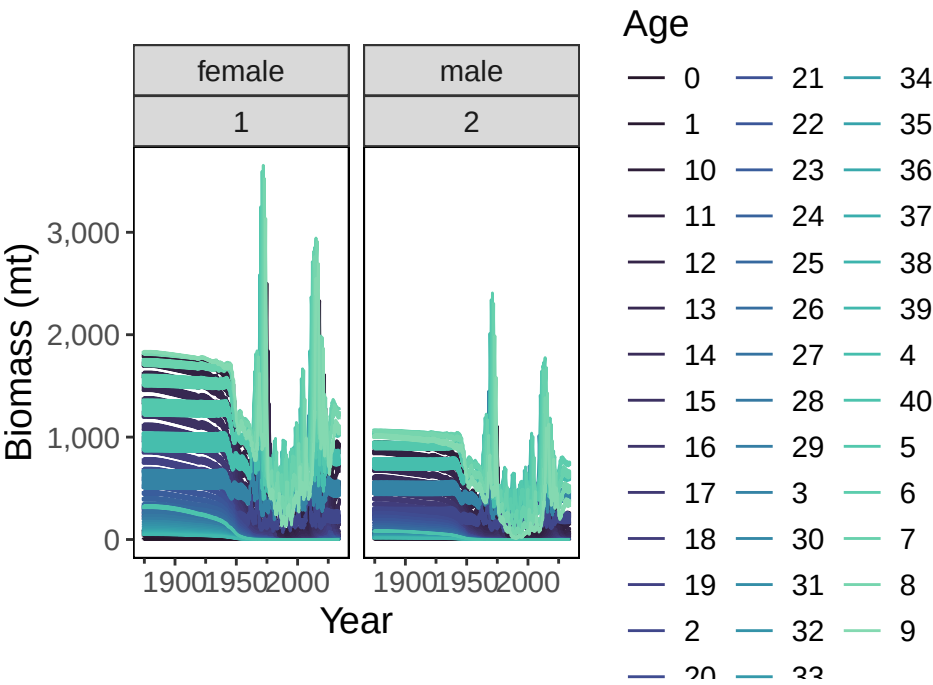


Figure 3: Biomass (B) time series. The horizontal dashed line represents the limit reference point (msy mt).

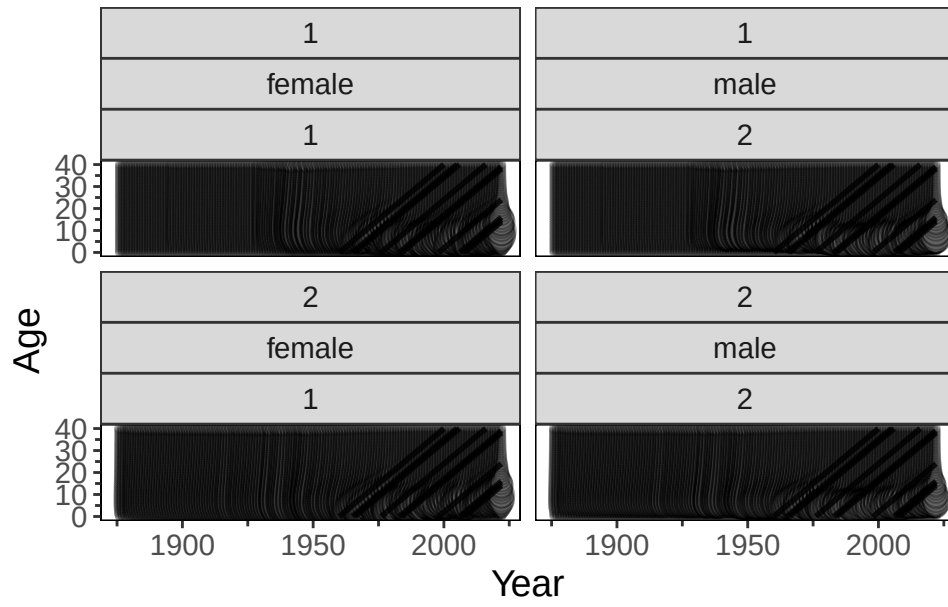


Figure 4: Fishery age composition (1876-2022). The area of the circle is proportional to the catch. Diagonal lines indicated the top 5\% strongest year classes.

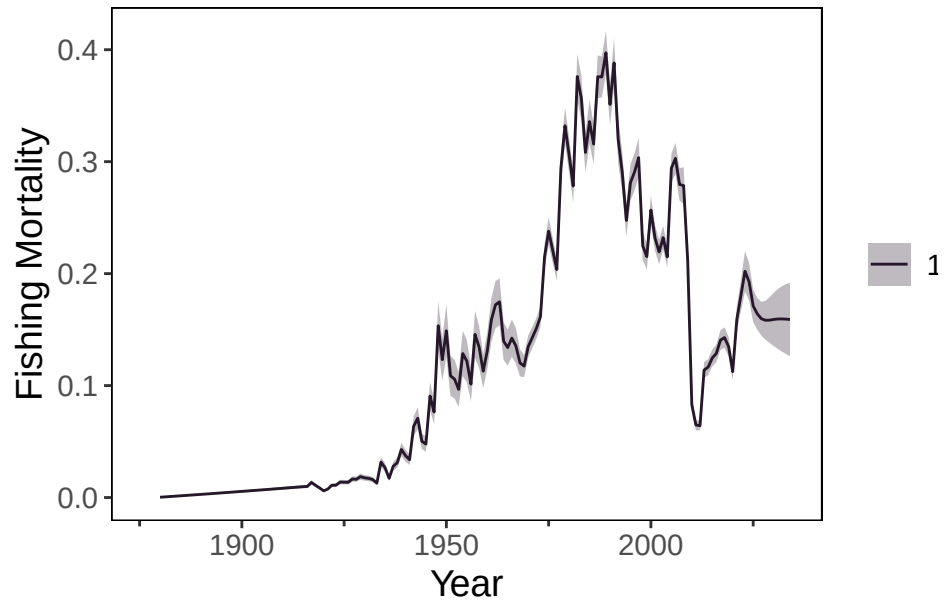


Figure 5: Fishing mortality (F) over time. The horizontal dashed line represents the target reference point (msy).

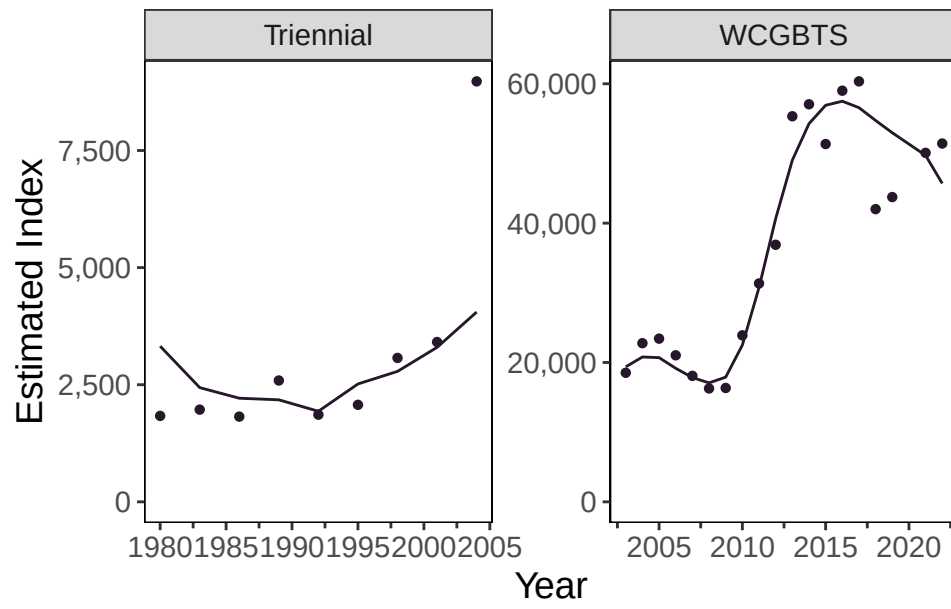


Figure 6: Catch per unit effort (CPUE) over time for fleet or survey. 95\% confidence intervals are shown for each survey/fleet.

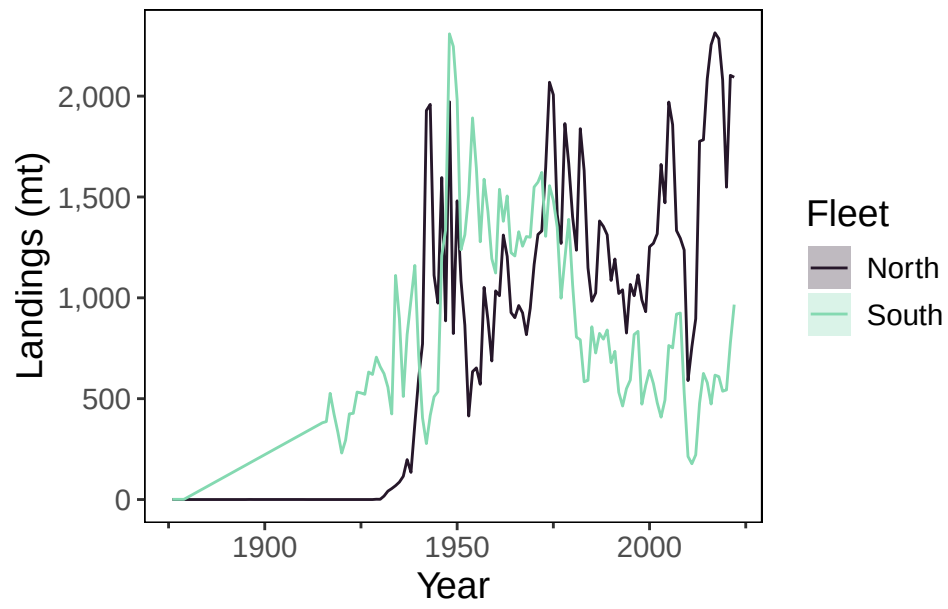


Figure 7: Historical landings by fleet.

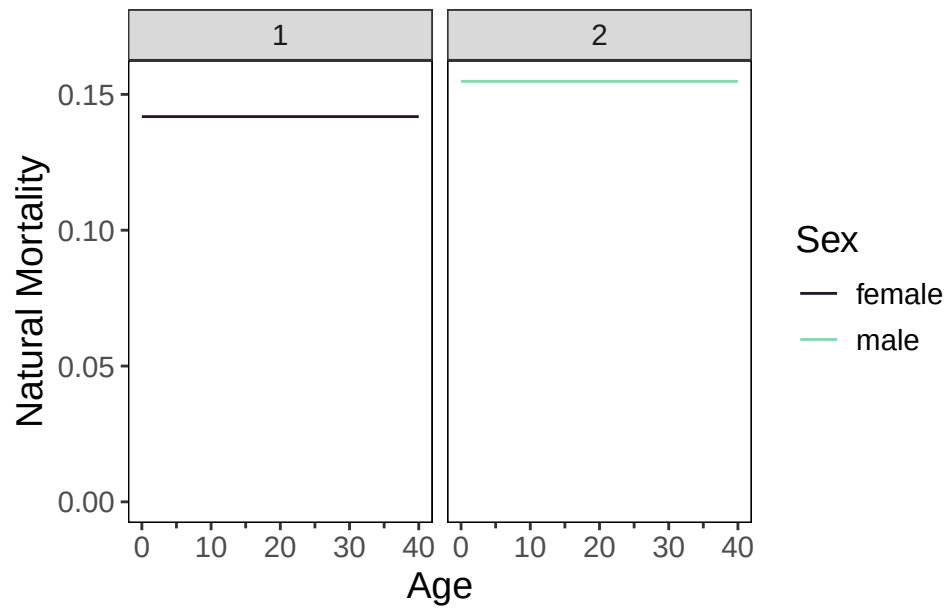


Figure 8: Natural mortality (M) for each age.

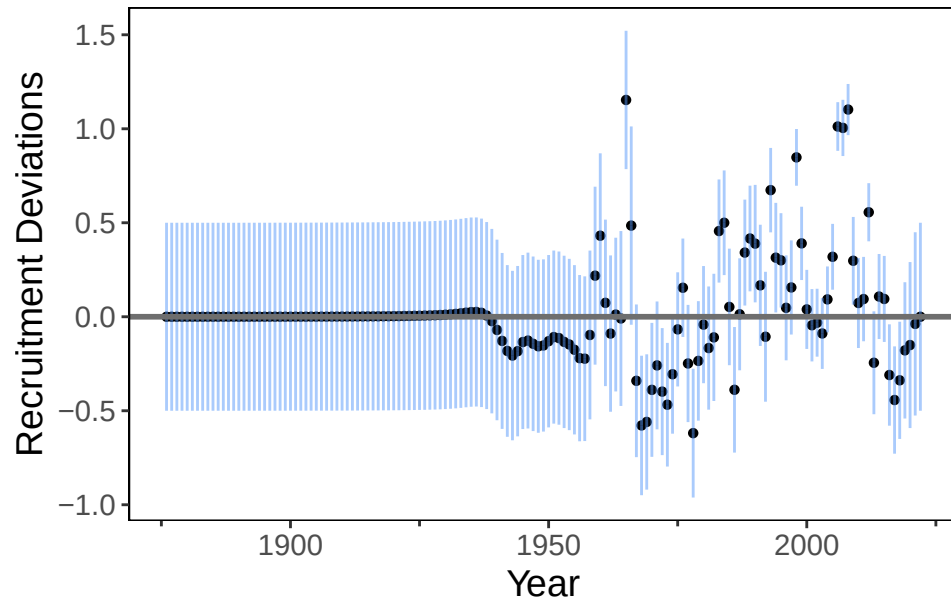


Figure 9: Annual deviations (on natural log scale) in the number of newly recruited fish the model estimates each year.

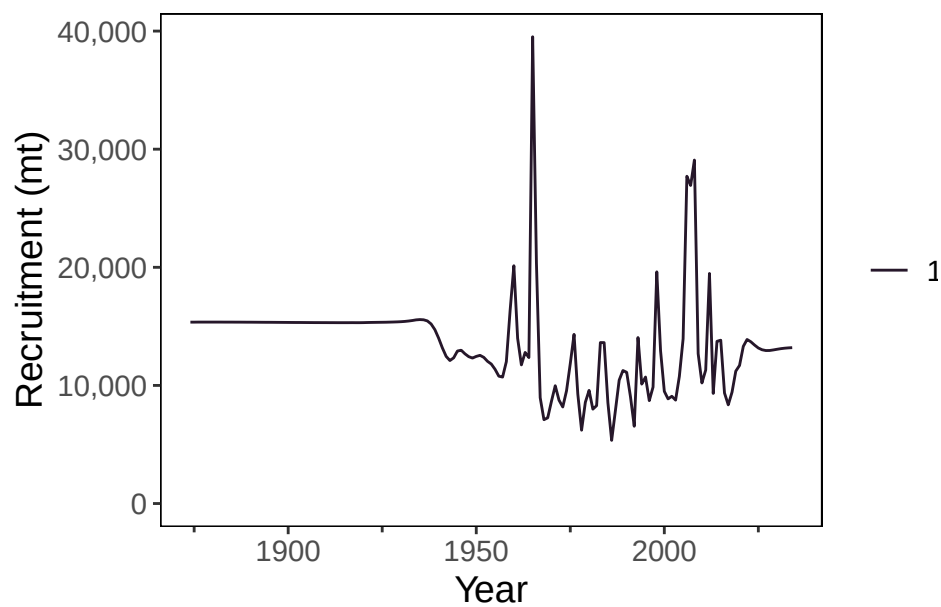


Figure 10: Estimated recruitment by the assessment model each year in mt.

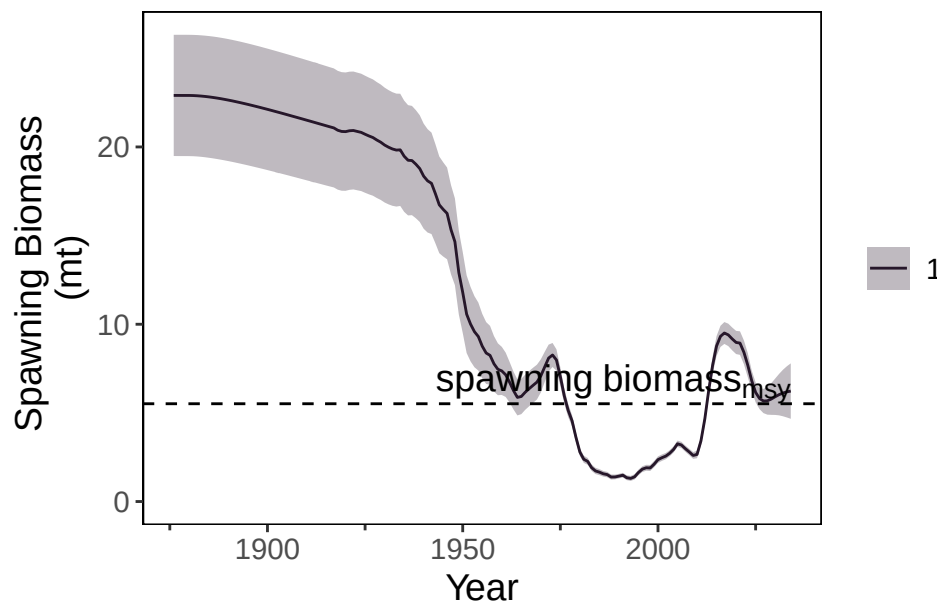


Figure 11: Estimated spawning biomass (SB) time series. The horizontal dashed line represents the spawning biomass associated with the biomass limit reference point (msy mt).

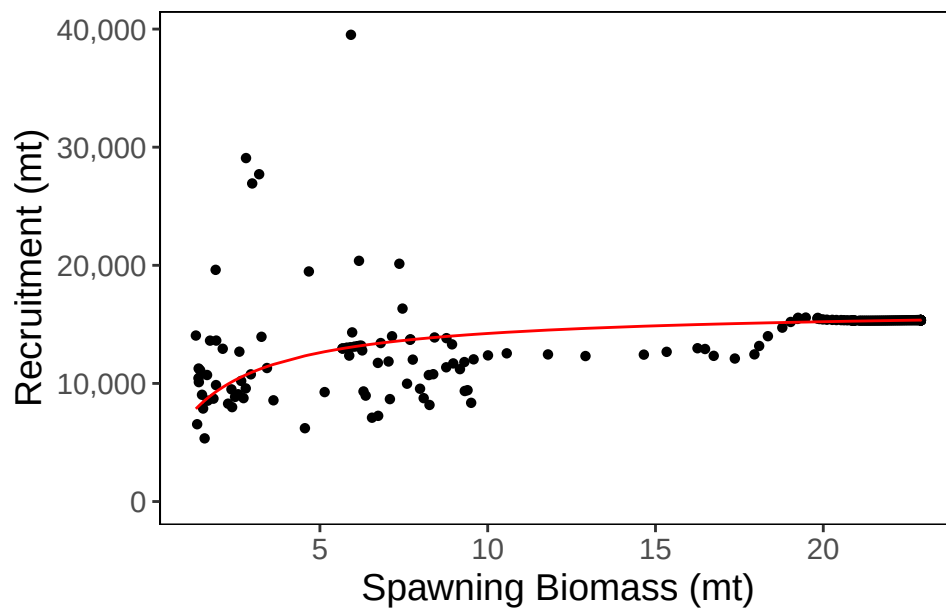


Figure 12: Stock recruitment relationship estimated by the assessment model.